Response to Reviewer 1

Manuscript **TP-1083** — *The Incommensurability Principle in Biological Transport*
Transport Phenomena (De Gruyter Brill)

We thank Reviewer 1 for the careful reading and for a constructive observation that has materially improved the framing of the paper. Below we reproduce the comment (paraphrased faithfully from the report) and detail our response. **All changes prompted by Reviewer 1 are typeset in blue in the highlighted copy of the manuscript.**

Comment 1 — Opening contrast (mouse vs. whale heart rate).

> *Reviewer comment.*
> *The introduction motivates the problem by contrasting the mouse (∼600 bpm) and the whale (∼20 bpm) heart rates. Heart rate alone is a misleading basis for the comparison; the relevant quantity is the volume of blood each heart pumps (cardiac output / stroke volume). The argument on the Womersley number should be reformulated accordingly and checked for consistency.*

Response. We agree, and we are grateful for this correction — it sharpens the physical logic and, in fact, repairs a potential inversion of the intended point. Because the Womersley number $\mathrm{Wo} = R\sqrt{\omega/\nu}$ *increases* with frequency, a comparison on heart rate alone would make the mouse appear *more* inertial than the whale, the opposite of our thesis.

We have rewritten the opening contrast in terms of the pumped volume. Cardiac output scales as $\dot{Q} \propto M^{3/4}$ and stroke volume as $\propto M$, so a single whale contraction ejects a blood volume larger than the mouse's by about seven orders of magnitude. That volume is carried by vessels whose radius scales as $R \propto M^{3/8}$, and it is the radius—not the rapid murine heart rate—that dominates the Womersley number. Combining the allometric scalings $R \propto M^{3/8}$ and $f \propto M^{-1/4}$ gives

$$\mathrm{Wo}_0 \;\propto\; M^{3/8}\sqrt{M^{-1/4}} \;=\; M^{3/8}\,M^{-1/8} \;=\; M^{1/4},$$

i.e. the Womersley number *rises* with body mass despite the slower large-animal heart beat. This resolves precisely the apparent paradox the reviewer identified.

Implementation and verification.

- The revision is *additive*: the original sentence is preserved and the new, rigorous framing is inserted (blue), so no text is silently removed.
- A cross-reference was added to the identical derivation already present in Section 6 (`sec:womersley_minimax`
- Internal-consistency check: the $M^{1/4}$ law predicts a human/mouse Womersley ratio of $(70000/20)^{1/4} \approx 7.7$, matching the Section 6 table (≈ 7.2) to within ∼7%.
- To avoid a residual ambiguity, a short author clarification (magenta in the highlighted copy) was added immediately afterward so that the mouse is *not* read as realising Murray's $\alpha = 3$ architecture: the mouse and whale cases are framed as the two single-mechanism limits (viscous α_t and wave $\alpha_w \approx 2$) of the unified Lagrangian, consistent with Sections 5–6 (where the mouse coronary is predicted in the transition range, not at $\alpha = 3$).

Location: Section 1 (Introduction), highlighted blue, with one adjoining magenta consistency clarification.

We thank the reviewer again: the opening is now both physically correct and more compelling.